Improving the Asymptotic Radiative Transfer Model to Better Characterize the Pure Snow Hyperspectral Bidirectional Reflectance

Anxin Ding, Shunlin Liang, *Fellow, IEEE*, Ziti Jiao, Han Ma, Alexander A. Kokhanovsky, and Jouni Peltoniemi

***Abstract*— The asymptotic radiative transfer (ART) model has been widely used in snow remote sensing. However, the anisotropic effects of snow reflectance challenge this model because of its underestimation in the forward-scattering direction. To exhibit these strong scattering properties of the snow surface, a microfacet specular kernel has been supplemented with the ART model (hereinafter named the ARTS model). In this study, we propose a method of multiplying by a correction term for improving the ART model (hereinafter named the ARTF model). We validate the performance of the ARTF model using various data sources. Our results demonstrate that: 1) the ARTF model has higher accuracy in characterizing snow bidirectional signatures, with R^2 and root mean square error (RMSE) values in the ranges from 0.722 to 0.990 and 0.007 to 0.041, respectively, than the ART (R^2 = 0.507–0.802 and RMSE = 0.038–0.088) and ARTS (R^2 = 0.686–0.962 and RMSE = 0.021–0.044) models, especially in the long-wave near-infrared region and 2) the ARTF model can effectively represent snow hyperspectral reflectance, while the ART and ARTS models significantly underestimate snow reflectance in the visible and shortwave near-infrared region. The R^2 values of these three models reach ∼0.99, and the RMSE values of the ARTF model range from 0.012 to 0.024, which are smaller than those of the ART (RMSE = 0.021–0.061) and ARTS (RMSE = 0.021–0.049) models. These results demonstrate that the ARTF model is better than the ART and ARTS models for characterizing snow hyperspectral bidirectional reflectance.**

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***Index Terms*— Asymptotic radiative transfer (ART) model, bidirectional reflectance distribution function (BRDF), cloud absorption radiometer (CAR), forward-scattering effect, hyperspectral reflectance, POLarization and Directionality of the Earth's Reflectances (POLDER), snow.**

I. Introduction

SNOW plays a key role in the global radiation balance, climate system, and water cycle. In particular, snow permanently covers Antarctica and has significant effects in the Arctic, which has warmed more rapidly than other regions of the planet during recent decades [1]–[7]. This phenomenon is known as Arctic amplification (AA) [3], [8]. The reflectance of the snow surface is far from isotropic and shows larger reflectance values in the forward rather than backward direction [1], [6], [9]–[14], which is opposite to other land surfaces. This reflectance anisotropy can be quantified by the bidirectional reflectance distribution function (BRDF) [1], [6], [9]–[14]. The high reflectance and large areal coverage of snow make it an essential determinant of Earth's radiation balance [4], [18], [19]. Therefore, it is greatly essential to understand the scattering properties and retrieve valuable information about snow for remote sensing applications.

Satellite remote sensing offers practical means to monitor snow changes in mid- to high-latitude regions [3], [5], [7], [21], [22], but it can generally obtain reflected radiation from snow surfaces in one particular direction and several wavelength bands [5], [18], [23], [24]. Therefore, accurate knowledge of angular and spectral reflectance characteristics is required to understand snow scattering properties and validate snow models [4]. Multidirectional measurements from aircraft can provide critical prior knowledge for understanding snow scattering signatures [25]. Field measurements can provide a unique remote sensing perspective for understanding snow angular and spectral patterns, which can be used to evaluate the ability of analytical models and develop BRDF models [4], [6], [7], [11], [21], [26], [27]. Peltoniemi *et al.* [6] directly analyzed the variability in reflectance patterns and albedo in relation to snow texture. Painter [28] used high-accuracy field measurements to explain snow anisotropic reflectance. Field measurements also have the potential to be used to predict effective spectrum wavelengths for particular applications [4]. Thus, it is necessary to perform a systematic investigation to improve the retrieval of snow information using satellite, aircraft, and ground-based angular and spectral signals.

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Snow modeling is also a key technique to understand snow scattering patterns [11]. A common method for modeling snow involves approximations of the radiative transfer equation or variants thereof, which simplifies multiple scattering based on the two-stream approximation [29], such as the Wiscombe and Warren (WW) [29] and asymptotic radiative transfer (ART) models [1], [2], [31]–[33]. Unlike these radiative transfer approximations, numerical methods are used to determine bidirectional reflectance quantities, and these numerical methods provide numerous computational streams for the calculation of angular radiance quantities, such as discrete ordinate radiative transfer (DISORT) models [34]. These models often assume that the snow particles are spherical in shape, which is inconsistent with the irregular shape of actual snow particles. Another modeling method uses a Monte Carlo ray-tracing scheme to compute a solution to the radiative transfer equation, which tends to provide an effective method for further examining and validating other snow reflectance models [12], such as the bicontinuous photon tracking (bic-PT) model [35]. A widely used modeling method applies semiempirical models to characterize the snow scattering characteristics, and these methods are developed by approximating and simplifying physical models. These semiempirical methods, such as kernel-driven BRDF models [11], [13], [15], [16], [19], [36], [37], maintain a certain physical justification and have the advantage of being easy to use. However, these surface BRDF models were developed for vegetation-soil systems and cannot perform well for characterizing snow BRDF signatures [13], [19].

The ART model for snow optical properties has been developed based on the semi-infinite, weakly absorbing media assumption [1], [2], [9], [11]. Validation tests have shown that this model exhibits a high accuracy within a view zenith angle (VZA) $< \sim 60°$ at a given solar zenith angle (SZA) [11], [33], and the ART model has been extensively applied to estimate snow parameters (e.g., snow particle size and snow albedo). Qu *et al.* [19] employed the ART model to simulate the snow BRDF database as *a priori* knowledge to retrieve the shortwave albedo in the Arctic region; Shi *et al.* [38] utilized the ART model to simulate the bidirectional reflection factors (BRFs) of snow for multiparameter estimation from Moderate Resolution Imaging Spectroradiometer (MODIS) observations. Recently, the ART model has been used to retrieve snow grain size and the absorption coefficient of pollutants [1], [32]. However, recent studies have recognized that the ART model underestimated the observed reflectance of snow at small scattering angles, particularly in the forward-scattering direction [4], [11], [13], [33]. To solve this problem, Lv *et al.* [4] added a specular kernel to the ART model to counter the forward and backward scattering of snow optical properties (hereinafter named the ARTS model). Jiao *et al.* [11] corrected the ART model for its use as a snow kernel in the kernel-driven model. In this study, we propose a method to revised the correction term of snow kernel based on the mathematical fitting method and multiplied this correction with the ART model to improve the ability of the ART model to characterize snow angular and spectral scattering signatures (hereinafter named the ARTF model), which can be applied to characterize snow reflective properties and incorporated into radiative transfer models [38]–[41]. The framework of this article is organized as follows. A description of the various snow multiangle and hyperspectral data sources and improvement of the ART model is presented in Section II. The validation results of the corrected ART model are demonstrated and analyzed in Section III. A discussion of the corrected ART model and uncertainties for snow data is presented in Section IV. Finally, we describe the primary conclusions of this study in Section V.

II. Models and Data

A. ART Model

The ART model assumes a layer of semi-infinite, weakly absorbing, and plane-parallel media and was proposed by Kokhanovsky and Zege [33]. The ART model is analytical and easily inverted against a set of multiangle observations and simulated hyperspectral reflectance. The equations can be expressed as follows:

$$R(\theta_s, \theta_v, \varphi) = R_0(\theta_s, \theta_v, \varphi) \cdot e^{-y \cdot K_0(\theta_s) K_0(\theta_v)/R_0(\theta_s, \theta_v, \varphi)} \tag{1}$$

$$R_0(\theta_s, \theta_v, \varphi) = \frac{A + B(\cos\theta_s + \cos\theta_v) + C\cos\theta_s\cos\theta_v + P(\xi)}{4(\cos\theta_s + \cos\theta_v)} \tag{2}$$

$$P(\xi) = 11.1e^{-0.087(180-\xi)} + 1.1e^{-0.014(180-\xi)} \tag{3}$$

$$\cos\xi = \cos\theta_s\cos\theta_v + \sin\theta_s\sin\theta_v\cos\varphi \tag{4}$$

$$K_0(\theta) = \frac{3}{5}\cos\theta + \frac{1+\sqrt{\cos\theta}}{3} \tag{5}$$

$$y = \sqrt{4\pi L(\chi_{\text{ice}} + M)/\lambda} \tag{6}$$

where $R(\theta_s, \theta_v, \varphi)$ is the bidirectional reflectance or the reflection function of a semi-infinite snow layer and $R_0(\theta_s, \theta_v, \varphi)$ represents the surface reflectance of a semi-infinite snow layer at zero absorption. The function $K_0(\theta)$ is the escape function in the radiative transfer theory, which determines the angular distribution of light escaping from the semi-infinite nonabsorbing medium. The values of A, B, and C equal 1.247, 1.186, and 5.157, respectively. $P(\xi)$ is the phase function, which determines the angular distribution of light. y is mainly related to the optical and structural properties of snow particles (e.g., χ_{ice} and L). χ_{ice} is the imaginary part of the ice refractive index [42], and λ is the wavelength. The value of L is proportional to the effective snow grain size, and M is directly proportional to the mass concentration of pollutants in snow. In this study, we only use pure snow data, so the value of M is approximately equal to 0. θ_s, θ_v, and φ represent the SZA, VZA, and relative azimuth angle (RAA), respectively. Note that ξ in (3) represents the phase angle in degrees.

B. Improved ART Model

1) ARTS Model: Considering that the ART model tends to underestimate the reflectance in the forward-scattering direction, Lv *et al.* [4] use a microfacet specular kernel proposed by Beckmann and Spizzichino [43] to counter the reflectance

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from snow surfaces. The equations for the ARTS model are given as follows:

$$R_{\mathrm{ARTS}}(\theta_s,\theta_v,\varphi) = R(\theta_s,\theta_v,\varphi) + k_s R_s(\theta_s,\theta_v,\varphi) \tag{7}$$

$$R_s(\theta_s,\theta_v,\varphi) = \frac{1}{4\cos\theta_s\cos\theta_v}\cdot F(\xi,n)G(\theta_s,\theta_v,\varphi) \cdot\frac{P(\beta,\delta)}{P_n} \tag{8}$$

$$P(\beta,\delta) = \frac{P_n}{\pi\delta^2\cos^4(\beta)}\cdot e^{-\frac{\tan^2(\beta)}{\delta^2}} \tag{9}$$

$$\cos\beta = \frac{\cos\theta_s+\cos\theta_v}{2\cos\frac{\xi}{2}} \tag{10}$$

where k_s is a coefficient parameter to constrain the specular portion during the modeling process, $R_s(\theta_s,\theta_v,\varphi)$ is the specular kernel, δ is the roughness parameter, and P_n is the normalization factor, which is used to determine the probability of a microfacet appearance. Parameter G, which accounts for the shadowing and masking of one facet, is defined as follows:

$$G(\theta_s,\theta_v,\varphi) = \min(1, E_1, E_2) \tag{11}$$

$$E_1 = \frac{\cos\beta\cos\theta_v}{\cos\frac{\xi}{2}},\ E_2 = \frac{\cos\beta\cos\theta_s}{\cos\frac{\xi}{2}} \tag{12}$$

$$F(\xi,n) = 0.5\left(\frac{g-\cos\frac{\xi}{2}}{g+\cos\frac{\xi}{2}}\right)^2 \times\left[1+\left(\frac{\left(g+\cos\frac{\xi}{2}\right)\cos\frac{\xi}{2}-1}{\left(g-\cos\frac{\xi}{2}\right)\cos\frac{\xi}{2}+1}\right)^2\right] \tag{13}$$

$$g = \cos^2\frac{\xi}{2}+n^2-1 \tag{14}$$

where $F(\xi, n)$ represents the Fresnel factor, which drives the amount of light reflected at the boundary between air and snow. β, E_1, and E_2 are the functions of θ_s, θ_v, and φ, which determine the angular distribution of light. n represents the refractive index of ice.

2) ARTF Model: The ART model does not account for the specular reflection of horizontally oriented ice crystals at the snow top. Therefore, the anisotropic effects of the ART model have a significant underestimation in the forward-scattering direction. To correct the underestimation of the reflectance in the forward-scattering direction, particularly at small scattering angles in the principal plane (PP) containing the directions of the incidence, observation, and normal angles to the snow layer [11], [33], [44], Jiao *et al.* [11] propose a correction term to strengthen the forward-scattering effect of snow cover, which has been used in a kernel-driven BRDF model framework to better model the anisotropic reflectance of pure snow. The correction form can be expressed as follows:

$$H(\xi,b) = 1-b\cdot\cos\xi\cdot e^{-\cos\xi}. \tag{15}$$

In order not to change the BRDF shape of the ART model and improve the accuracy of the corrected ART model during the correction process, when $\cos\xi = 1$ and $\cos\xi \times e^{-\cos\xi} = e^{-1}$, we revise (15) and introduce a parameter to (15). The equation can be expressed as follows:

$$H(\xi,b,a) = a-b\cdot\left(\cos\xi\cdot e^{-\cos\xi}-e^{-1}\right). \tag{16}$$

Then, we use H(1) and H(-1) to represent a and b parameters. Equation (16) can be further revised as follows:

$$H(z) = a-b[h(z)-h(1)] \tag{17}$$

$$h(z) = z\cdot\exp(-z),\quad z=\cos\xi. \tag{18}$$

When $\cos\xi = 1$, one can see that the parameter a is the correction term in the exact backward direction (the phase angle is equal to 0°)

$$H(1) = a. \tag{19}$$

When $\cos\xi = -1$, this is in the exact forward direction (the phase angle is equal to 180°)

$$H(-1) = a-b(h(-1)-h(1)). \tag{20}$$

It is clear that H(1) and H(-1) have a clear physical meaning; they presented the angular distribution of light and control the relative amount of forward and backward, which can account for the specular reflectance of the snow surface.

We use (19) and (20); the parameter b can be expressed as (21). It is proportional to the difference in correction factors in the forward and backward directions

$$b = \frac{H(1)-H(-1)}{h(-1)-h(1)}. \tag{21}$$

In this study, we multiplied (17) with the ART model. The equation for the ARTF model is given as follows:

$$R_{\mathrm{ARTF}}(\theta_s,\theta_v,\varphi) = R(\theta_s,\theta_v,\varphi)\cdot H(z). \tag{22}$$

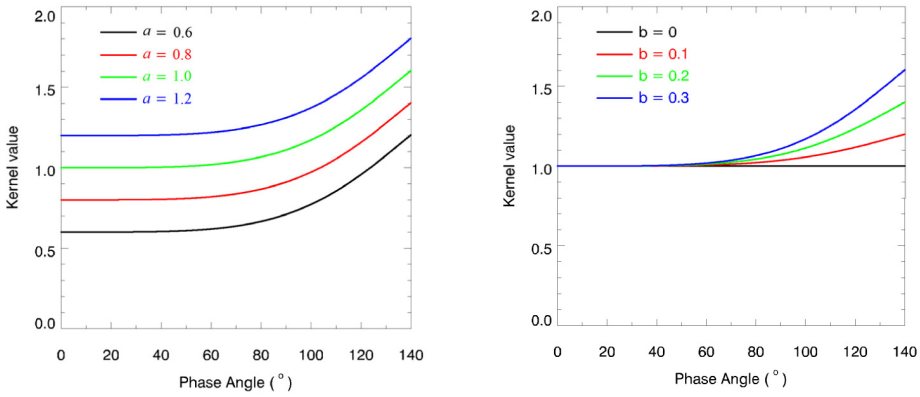
One can see that this snow reflectance model is much simpler than that given by (7).

We analyzed the variation of the correction term with the phase angle. Fig. 1 shows the kernel value of the correction term under different a and b values. It is clear that the a value controls the magnitude of the snow reflectance, which slightly adjusts the correction term to better fit the scattering characteristics of snow. The correction effect becomes more significant for the forward-scattering direction with an increasing b value and has less effect on the backward-scattering direction. At $a = 1$ and $b = 0$, the ARTF model will be degraded to the original format of the ART model. Therefore, we use this method to address the underestimation of the ART model in the forward-scattering direction.

C. Data

To validate the performance of the corrected ARTF model with respect to snow hyperspectral bidirectional reflectance, we used a variety of directional and spectral snow reflectance datasets, including the POLarization and Directionality of the Earth's Reflectances (POLDER) BRDF database, cloud absorption radiometer (CAR) data, field-measured hyperspectral multiangular data, and simulation data of the bic-PT model.

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(a) (b)

Fig. 1. Variation of the correction term with the phase angle under different (a) a and (b) b values. The value of b is fixed 0.3 in Fig.1(a), and the value of a is fixed 1.0 in Fig.1(b).

1) POLDER Data: The POLDER instrument, which provides multiangle and multispectral observations at a global scale, was used to measure the directionality and polarization of light reflected from natural targets. We use monthly POLDER data, and the variability of snow conditions may influence the variation in the BRDF signatures during the one-month collection period. To ensure homogeneity of the pixels from the POLDER BRDF databases, we only selected multiangle snow data with a coverage of 100%. We further removed data with an average reflectance < 0.4 in the blue band (0.490 μm) because the pure snow surface should have a high reflectance in the blue band. Finally, we made use of 864 pixels in each band for our study, which is detailed by Ding *et al.* [13].

2) CAR Data: The CAR is an airborne multiwavelength scanning radiometer designed to operate from various aircraft platforms. It has a unique design with a wide field of view (190°) and a small instantaneous field of view (1°), which allows it to scan in a plane perpendicular to the direction of flight. These measurements have been used to derive the spectral and angular distribution of reflectance at the full range of zenith and azimuth angles, and CAR data are valuable for assessing the accuracy of BRDF models [3], [22], [25].

3) Field-Measured Data: The field-measured data were obtained using the Finnish Geodetic Institute's Field Goniospectropolariradiometer (FIGIFIGO) with a spectral range of 350–2500 nm [6], [21], [26]–[28]. The interval of the spectral sampling is full-width at half-maximum of 0.003 μm from 0.4 to 1 μm and 0.01 μm from 1 to 2.5 μm. The VZA range was ±80° from nadir, and the zenith angular resolution was approximately 5°. In addition, we removed the low-quality spectral data and used 61 sets of data for model validation [12].

4) Simulation Data: The bic-PT model was proposed by Xiong and Shi [35] based on the Monte Carlo ray-tracing technique and can simulate snow directional and spectral data. The bic-PT model considers snow as a bicontinuous medium, which agrees reasonably well with the real snow microstructure. For the parameter settings listed in Table I, we referred to related studies. These simulation data can provide a point of reference independent of the satellite, aircraft, and ground data.

TABLE I
INPUT PARAMETERS OF THE BIC-PT MODEL

	Value (interval)	Unit
Monte Carlo superposition (N)	1000	--
photon total	50000	--
equivalent grain radius	0.2	mm
structure parameter (b)	5	--
snow density	0.3	g/cm^3
snow depth	1.5	m
wavelength number	211	--
wavelength	0.4-2.3 (0.01)	μm
SZA	0-70 (10)	degrees (°)
VZA	5-70 (5)	degrees (°)
RAA	1-359 (2)	degrees (°)
soil reflectance	0	--
stream number	32	--
order number	4	--

In this study, the root mean square error (RMSE), the mean relative error (MRE), and bias values, which are expressed in (23)–(25), respectively, were applied as the quality assessment indices

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{i=1}^{n}(y_2 - y_1)^2}{n - \mathrm{num}}} \tag{23}$$

$$\mathrm{MRE} = \frac{1}{n - \mathrm{num}} \sum_{i=1}^{n} \left| \frac{y_2 - y_1}{y_1} \right| \times 100\% \tag{24}$$

$$\mathrm{bias} = \frac{\sum_{i=1}^{n}(y_2 - y_1)}{n - \mathrm{num}} \tag{25}$$

where y_1 represents the POLDER or field-measured reflectance and y_2 represents the reflectance retrieved using the corrected ART model. num represents the number of parameters in the snow model.

III. RESULTS AND ANALYSIS

A. Results With POLDER Data

First, we validate the ability of the ARTF model to characterize snow BRDF signatures using POLDER data. Fig. 2 shows the reconstructed BRDF shapes using three snow models in the PP in the red and shortwave near-infrared (SW-NIR) bands (i.e., 0.67 and 1.02 μm, respectively). The results indicate that these three snow models can generally

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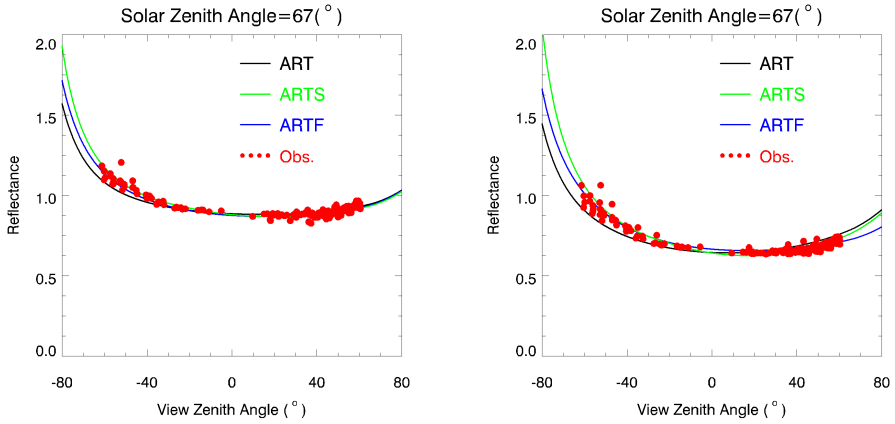


Fig. 2. BRDF shapes reconstructed by the ART (black), ARTS (green), and ARTF (blue) models in the PP compared with POLDER observations (red dots). The center wavelengths are (a) red and (b) SW-NIR (i.e., 0.67 and 1.02 μm, respectively). The negative axis along the PP is the forward-scattering direction, and the positive axis is the backward-scattering direction.

TABLE II

BRDF PARAMETERS AND STATISTICAL RESULTS OF THE THREE MODELS FOR POLDER DATA IN THE RED AND NEAR-INFRARED (SW-NIR) BANDS

Model	Bands	Parameters			Forward direction			Backward direction			All		
		L,cm	δ/a	k_s/b	R^2	RMSE	Bias	R^2	RMSE	Bias	R^2	RMSE	Bias
ART	red	0.45	--	--	0.867	0.027	-0.012	0.937	0.020	0.007	0.895	0.026	0.000
	SW-NIR	0.42	--	--	0.865	0.032	0.001	0.849	0.045	0.023	0.800	0.041	0.002
ARTS	red	1.02	0.50	0.10	0.940	0.019	0.004	0.960	0.020	0.005	0.924	0.019	-0.001
	SW-NIR	0.55	0.50	0.20	0.956	0.020	0.010	0.953	0.018	0.004	0.926	0.024	0.002
ARTF	red	1.02	1.00	0.05	0.917	0.018	0.005	0.920	0.020	-0.005	0.916	0.020	-0.001
	SW-NIR	0.05	0.79	0.12	0.956	0.020	0.002	0.923	0.023	-0.002	0.929	0.023	-0.001

reconstruct snow BRDF shapes. However, the BRDF shapes reconstructed by the ART model exhibit significant underestimation in the forward direction, and the ART model tends to overestimate in the backward direction in the SW-NIR band. In contrast, the ARTS and ARTF models perform better than the ART model in reconstructing snow BRDF shapes. Although the BRDF shapes reconstructed by the ARTS and ARTF models have little difference in the red band, the ARTS model somewhat overestimates in the forward direction in the SW-NIR band relative to the POLDER observations. Table II shows the BRDF parameters and statistical results of the three models for the POLDER pixel. We also counted the fitting accuracy of these three snow models in the forward and backward directions, which has a similar conclusion to the result of Fig. 2. The ART model can generally fit POLDER data with R^2 values of 0.800–0.895 and RMSEs of 0.026–0.041. In comparison, the ARTS and ARTF ($R^2 = 0.915$–0.929 and RMSE $= 0.019$–0.024) models perform better than the ART model with the POLDER data. In addition, the results of the ARTF model ($R^2 = 0.929$ and RMSE $= 0.023$) are superior to those of the ARTS model ($R^2 = 0.926$ and RMSE $= 0.024$) in the SW-NIR band, and the ARTS model has an overcorrection problem for POLDER data.

Fig. 3 shows the scatter plots that compare all POLDER pixels with the two respective model-observation fits in the red and SW-NIR bands. The ARTF ($R^2 = 0.718$ and RMSE $= 0.033$) and ARTS ($R^2 = 0.683$ and RMSE $= 0.035$) models fit these POLDER BRFs with a higher accuracy than the ART ($R^2 = 0.612$ and RMSE $= 0.038$) model in the red band. The MREs of the ARTF and ARTS models are reduced by 0.4% and 0.2%, respectively, compared to that of the ART model. It is clear that the inversion accuracy of the ARTF model is slightly higher than that of the ARTS model for all POLDER data. The R^2 values of the ARTF ($R^2 = 0.789$) and ARTS ($R^2 = 0.754$) models are higher than that of the ART ($R^2 = 0.679$) model in the SW-NIR band, which indicates high correlations of the ARTF and ARTS models with the POLDER observations. The RMSE values of the ARTF (RMSE $= 0.041$) and ARTS (RMSE $= 0.044$) models are much smaller than that of the ART (RMSE $= 0.050$) model, and the MREs of the ARTF and ARTS models are reduced by 1.3% and 0.7%, respectively, compared to that of the ART model. In addition, the ART model significantly underestimates the POLDER observations at high reflectance but overestimates the POLDER observations at low reflectance in both the red and SW-NIR bands. These results demonstrate that the ARTF model performs better than the ART and ARTS models with respect to the characterization of snow BRDF signatures. Notably, all snow models present better results in the red band than in the SW-NIR band, which is probably due to increased uncertainties in the SW-NIR band caused by atmospheric correction.

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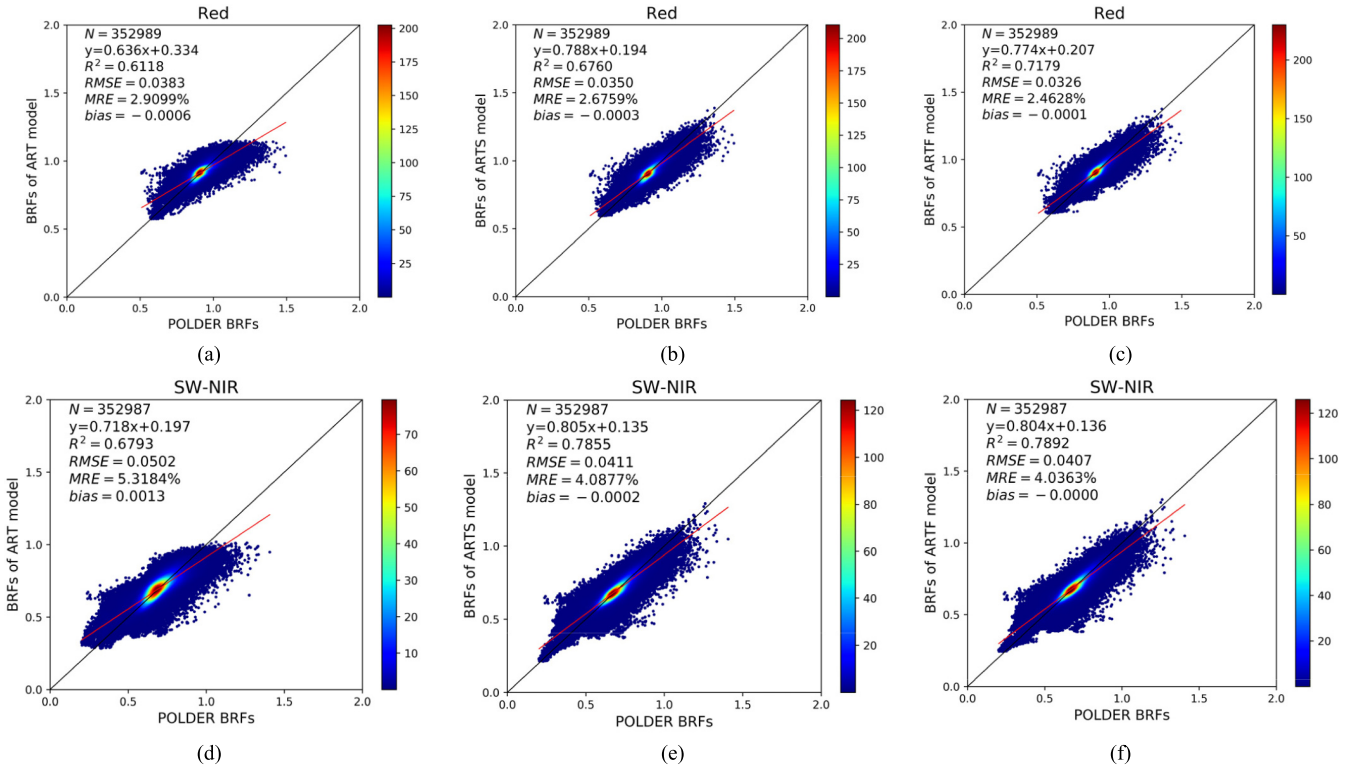


Fig. 3. Scatter plots that compare all POLDER observations with the model-observation fits by the ART, ARTS, and ARTF models from (Left) to (Right) in (a)–(c) red and (d)–(f) SW-NIR bands (i.e., 0.67 and 1.02 μm, respectively).

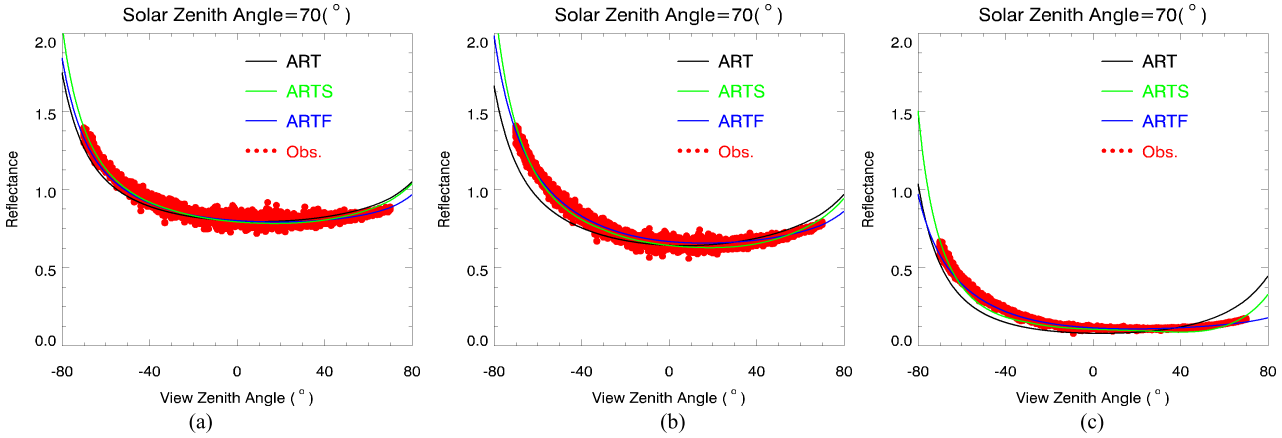


Fig. 4. BRDF shapes reconstructed by the ART (black), ARTS (green), and ARTF (blue) models in the PP compared with CAR observations (red dots). The wavelengths are (a) red, (b) SW-NIR, and (c) LW-NIR (i.e., 0.677, 1.032, and 1.649 μm, respectively). The negative axis along the PP is the forward-scattering direction, and the positive axis is the backward-scattering direction.

B. Results With CAR Data

To validate the results based on the POLDER data, we further assess the performances of the ARTF model using the CAR data. We also analyze the ability of the ARTF model in the long-wave near-infrared (LW-NIR) band (i.e., 1.649 μm), which has an important influence on ice crystal morphology [3].

Fig. 4 shows the BRDF shapes reconstructed by these three models for three bands. The conclusions obtained from these three models are similar to the validation results of POLDER data in the red and SW-NIR bands. Both the ARTF and ARTS models have high accuracy in characterizing the snow BRDF signatures, and the ART model has an underestimation in the forward direction and overestimation in the backward direction, especially in the SW-NIR band. The snow reflectance is significantly lower in the LW-NIR band than in the red and SW-NIR bands, and the LW-NIR band exhibits strong scattering characteristics in the forward direction and shows more isotropy in the backward direction. The ART model underestimates the forward direction and overestimates the backward direction, and the ARTS model agrees well with the CAR data for reconstructing snow BRDF shapes, but it slightly overestimates the forward and backward directions. In contrast, the ARTF model performed well with respect to the characterization of snow BRDF signatures.

Table III shows the BRDF parameters and statistical results of the three models. It is obvious that the ARTF model

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TABLE III
BRDF PARAMETERS AND STATISTICAL RESULTS OF THE THREE MODELS FOR THE CAR DATA IN THE RED, SW-NIR, AND LW-NIR BANDS

Model	Bands	Parameters			Forward direction			Backward direction			All		
		L,cm	δ/a	k_s/b	R^2	RMSE	Bias	R^2	RMSE	Bias	R^2	RMSE	Bias
ART	red	3.83	--	--	0.768	0.040	-0.043	0.759	0.045	0.013	0.725	0.046	0.007
	SW-NIR	0.38	--	--	0.790	0.050	-0.107	0.825	0.049	0.010	0.795	0.051	0.000
	LW-NIR	0.30	--	--	0.721	0.051	-0.070	0.710	0.046	0.015	0.640	0.052	-0.006
ARTS	red	5.99	0.60	0.12	0.865	0.043	0.009	0.764	0.036	-0.002	0.835	0.040	0.005
	SW-NIR	0.56	0.64	0.26	0.921	0.030	0.011	0.933	0.030	0.008	0.938	0.032	0.001
	LW-NIR	0.55	0.79	0.42	0.931	0.024	0.011	0.911	0.020	0.014	0.942	0.025	-0.006
ARTF	red	0.00	0.88	0.06	0.865	0.034	-0.010	0.765	0.028	-0.008	0.867	0.032	-0.001
	SW-NIR	0.01	0.76	0.13	0.964	0.025	-0.015	0.931	0.023	-0.012	0.954	0.024	-0.001
	LW-NIR	0.02	0.20	0.17	0.987	0.010	-0.009	0.974	0.008	-0.002	0.986	0.009	0.001

shows the highest accuracy in describing the snow BRDF features, followed by the ARTS model. The overall R^2 values of the ARTF model in the range of 0.867–0.986 indicate a high correlation with the CAR data. However, the R^2 values of the ART (ARTS) model are smaller than those of the ARTF model, ranging from 0.640 to 0.795 (0.835–0.942). The RMSE of the ARTF (RMSE = 0.009–0.032) model is much smaller than those of the ART (RMSE = 0.046–0.052) and ARTS (RMSE = 0.025–0.040) models. In general, the ARTF model performs better than the ARTS and ART models in characterizing snow BRDF signatures, especially in the LW-NIR band.

Fig. 5 compares the overall fitting results of the three models. It is obvious that these three snow models generally seem to be highly accurate in capturing CAR data. The ARTF model, which is highly consistent with the CAR data, fits these CAR BRFs with higher accuracy than the ART and ARTS models. The R^2 values of the ARTF model range from 0.898 to 0.990, indicating that the ARTF model fits these CAR data more accurately than the ARTS (R^2 = 0.851–0.946) or ART (R^2 = 0.549–0.797) models. The RMSEs of the ARTF model range from 0.007 to 0.025, which are much smaller than those of ARTS (RMSE = 0.021–0.030) and ART (RMSE = 0.040–0.049) models. The MREs of the ARTF model are reduced by 1.2%, 2.5%, and 21.2% (0.5%, 0.6%, and 5.4%) compared to those of the ART (ARTS) model in the red, SW-NIR, and LW-NIR bands, respectively. The ARTF model has significant improvements in capturing the forward-scattering effect compared with the ARTS and ART models, especially in the LW-NIR band. These results demonstrate that the ARTF model can fit the CAR data with a high degree of accuracy.

C. Results With Field-Measured Data

In this section, we validate the ability of the ARTF model with respect to the characterization of snow hyperspectral bidirectional reflectance using field-measured data. The conclusions obtained by these three models are similar to the validation results of POLDER and CAR data (please see the Appendix). Based on the overall R^2 and RMSE values, the ARTF and ARTS models perform better than the ART model for the three bands. However, the ARTS model shows higher accuracy than the ARTF model in the red and SW-NIR bands, which may be because the field-measured data exhibit strong forward-scattering effects compared with the POLDER and CAR data.

Then, we validate the ability of the ARTF model with respect to the characterization of snow hyperspectral reflectance using these data. Three observation directions, i.e., the forward, zenith, and backward directions, are selected. Fig. 6 shows the spectral reflectance of the field-measured data and the spectral reflectance fitted using the three models, which generally fits the spectrum changes and shows a high correlation with the field-measured data. However, the ART model significantly underestimates the spectral reflectance of the field-measured data in the forward direction in the visible and SW-NIR regions. The ARTS model has a slight overestimation in the forward direction in the LW-NIR region (wavelength =$\sim$1.5 and 2.0 μm), and this model overestimates the spectral reflectance of field-measured data in the zenith and backward directions in the visible and SW-NIR regions. In contrast, the ARTF model performs better than the ART and ARTS models with respect to the characterization of snow hyperspectral reflectance. Table IV shows the retrieved parameters and statistical results of the three models in three observation directions. The R^2 values of these three models reach 0.99, indicating a high correlation with field-measured data. The RMSEs of the ARTF model range from 0.018 to 0.024, which are smaller than those of the ARTS (RMSE = 0.025–0.031) and ART (RMSE = 0.025–0.059) models. In addition, the bias of the ART model can reach -0.029 in the forward direction, which demonstrates that the ART model yields a significant underestimation in the forward direction. Therefore, significant improvements in capturing snow hyperspectral bidirectional reflectance can be made by using the ARTF model for field-measured data.

Fig. 7 shows density scatter plots corresponding to the field-measured and simulated spectral reflectances using the three models. It is obvious that the ARTF model shows a

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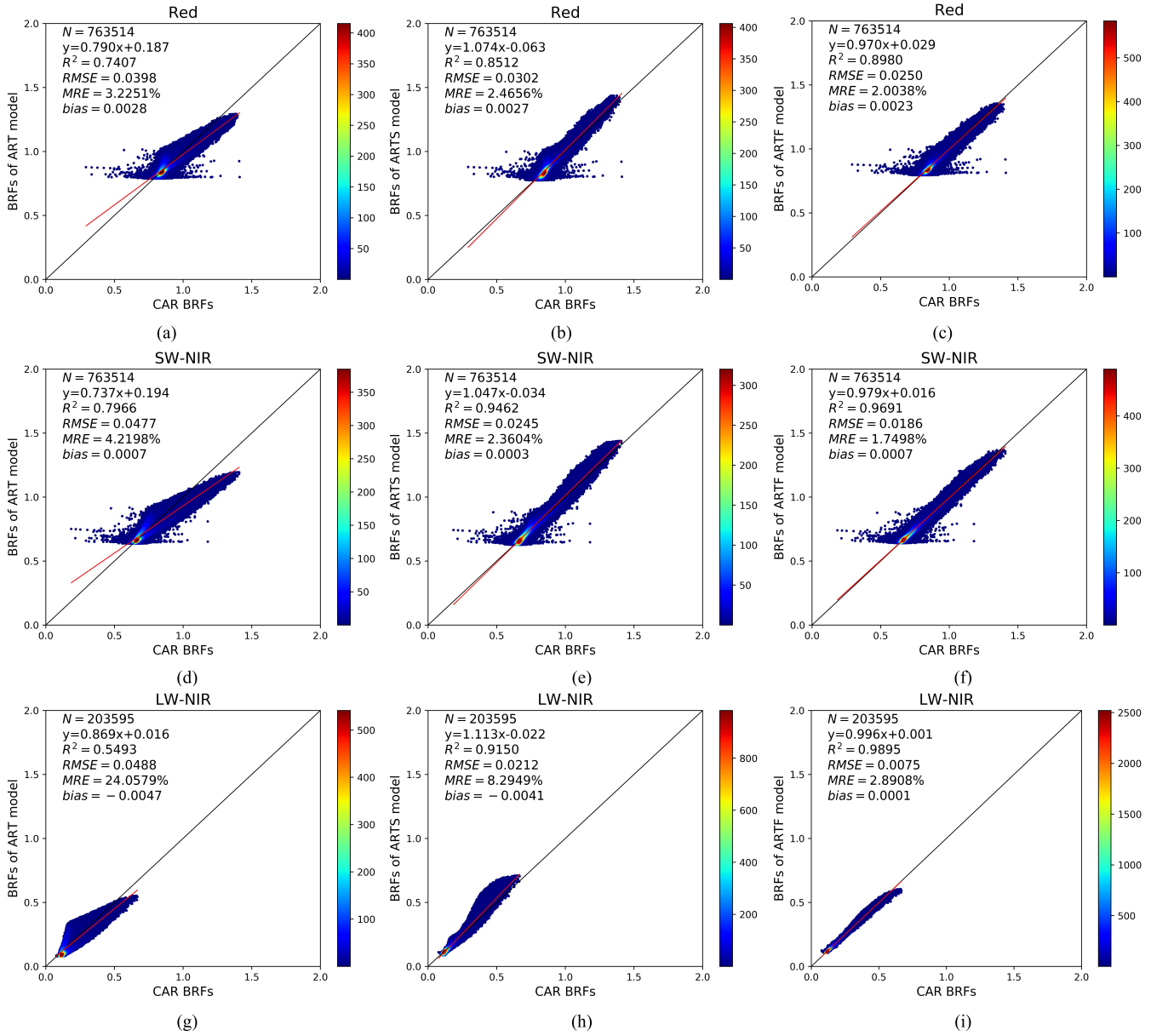


Fig. 5. Scatter plots that compare all CAR observations with the model-observation fits by the ART, ARTS, and ARTF models from (Left) to (Right) in (a)–(c) red, (d)–(f) SW-NIR, and (g)–(i) LW-NIR bands (i.e., 0.677, 1.032, and 1.649 μm, respectively).

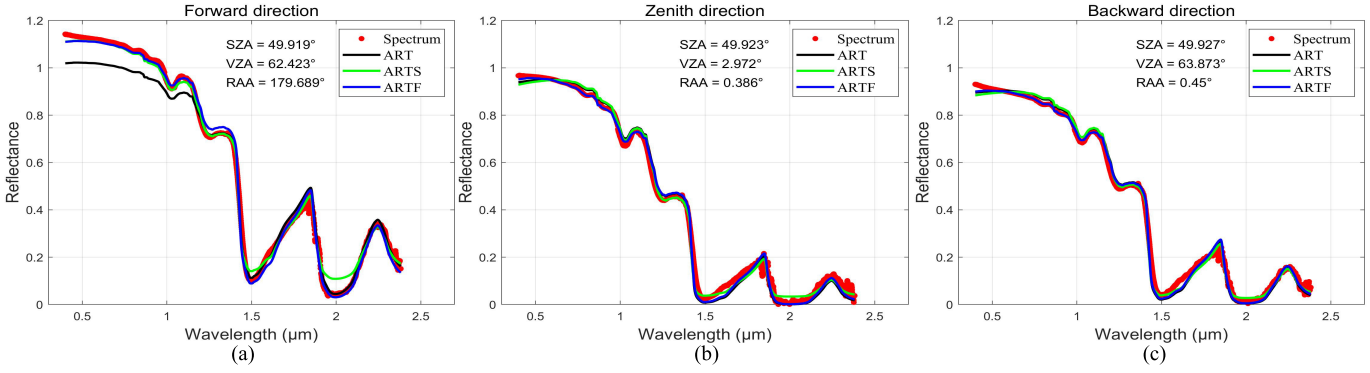


Fig. 6. Fitting of the spectral reflectance of the field-measured (red dots) data in three directions [i.e., (a) forward, (b) zenith, and (c) backward directions, from (Left) to (Right)] using the ART (black), ARTS (green), and ARTF (blue) models.

high correlation with the field-measured spectral reflectance ($R^2 = 0.997$), and the overall RMSE value is approximately 0.020. However, the overall R^2 values derived by the ART and ARTS models range from 0.993 to 0.995, and the RMSE results range from 0.026 to 0.031. In addition, the MRE of the ARTF model decreased by 3.2% and 3.4% compared to those

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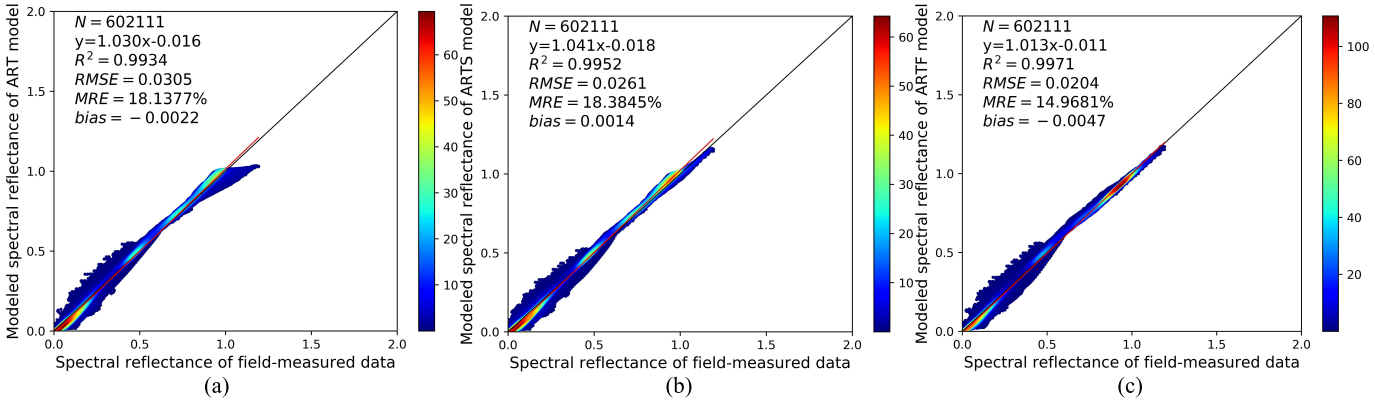


Fig. 7. Scatter plots that compare all field-measured spectral reflectance and spectral reflectance modeled from (Left) to (Right) by (a) ART, (b) ARTS, and (c) ARTF models.

TABLE IV

RETRIEVED PARAMETERS AND STATISTICAL RESULTS OF THREE MODELS FOR THE FIELD-MEASURED DATA IN THREE OBSERVATION DIRECTIONS' BANDS

Model	Directions	L,cm	δ	k_s	a	b	R^2	RMSE	Bias
ART	forward	0.16	--	--	--	--	0.998	0.059	-0.029
	zenith	0.31	--	--	--	--	0.998	0.026	-0.006
	backward	0.42	--	--	--	--	0.998	0.022	-0.002
ARTS	forward	0.32	0.97	0.67	--	--	0.997	0.026	0.000
	zenith	0.41	0.30	1.00	--	--	0.997	0.022	-0.001
	backward	0.42	1.00	0.00	--	--	0.998	0.025	0.002
ARTF	forward	0.21	--	--	1.10	0.30	0.998	0.024	-0.001
	zenith	0.27	--	--	0.95	0.30	0.999	0.021	-0.007
	backward	0.38	--	--	0.96	0.12	0.998	0.018	-0.004

of the ART and ARTS models, respectively. These results demonstrate that the ARTF model performs better than the ART and ARTS models with respect to the characterization of snow hyperspectral reflectance.

D. Results With Simulation Data of the Bic-PT Model

Finally, we further explore the ability of the ARTF model to characterize snow hyperspectral bidirectional reflectance using the bic-PT model, and these validation results are similar to the validation results of satellite–aircraft–ground data in characterizing snow BRDF signatures (please see the Appendix). Therefore, we focus on evaluating the ability of the ARTF model to simulate snow hyperspectral reflectance. Fig. 8 shows the spectral reflectance simulated using the bic-PT model and the results retrieved using these three snow models. The ARTF model can capture snow hyperspectral reflectance effectively, and the ART and ARTS models significantly underestimate the fitting of the hyperspectral reflectance simulated using the bic-PT model in the visible and SW-NIR regions, especially in the forward and backward directions. However, the spectral reflectance stimulated by these three snow models is slightly inconsistent with the reflectance simulated by the bic-PT model at a wavelength of ∼1.83 μm in the forward and backward directions, and the three snow models overestimated the results of the bic-PT model. In general, the ARTF model performed significantly better in characterizing the snow spectral reflectance simulated by the bic-PT model compared with the ART and ARTS models. Table V shows the fitting parameters and the statistical results obtained based on these three snow models. From this table, it is evident that the ARTF model maintained a high inversion accuracy, with the corresponding R^2 and RMSE values reaching ∼0.999 and ∼0.014, respectively, indicating a high correlation with the simulated data. However, the inversion accuracy of the ART (R^2 =∼0.999 and RMSE = ∼0.041) and ARTS (R^2 = ∼0.999 and RMSE =∼0.034) models decreased slightly in the visible and SW-NIR regions. In addition, the ARTS model slightly overestimated the fitting of the spectral reflectance simulated using the bic-PT model at wavelengths of ∼1.50 and ∼2.00 μm, which is consistent with the results reported in a previous study [4]. These results demonstrate that the ARTF model can fit the simulated hyperspectral reflectance with a high degree of accuracy.

IV. DISCUSSION

A. Snow Parameter Inversion

We have corrected the ART model to exhibit the strong scattering properties of the snow surface in the forward-scattering direction (i.e., ARTF model), and the difference may occur because the snow parameters retrieved by the ARTF differ from those retrieved by the ART model. Therefore, we use all POLDER data to explore the relationship between the snow parameters retrieved by the ART and ARTF models in the

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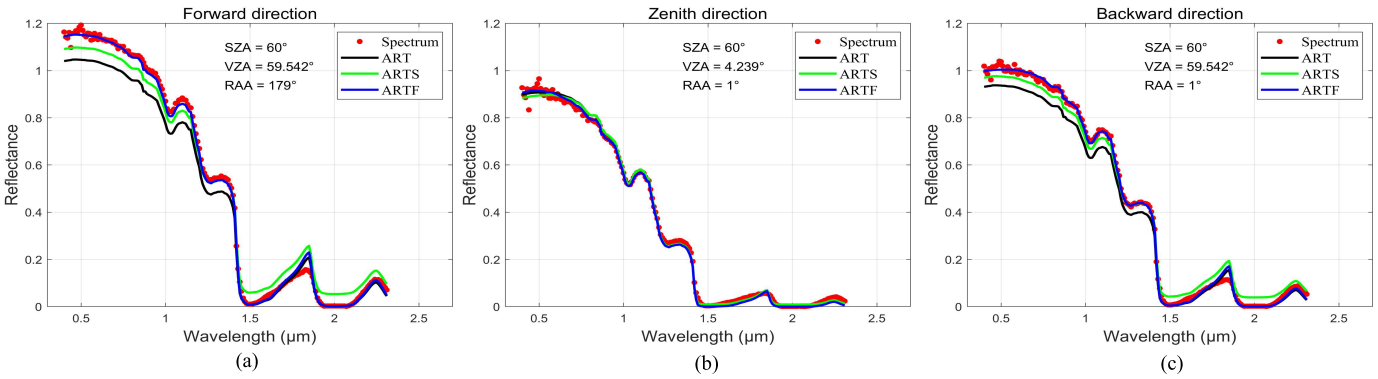


Fig. 8. Fitting of the spectral reflectance of the simulation data of the bic-PT model (red dots) in three directions [i.e., (a) forward, (b) zenith, and (c) backward directions from (Left) to (Right)] using the ART (black), ARTS (green), and ARTF (blue) models.

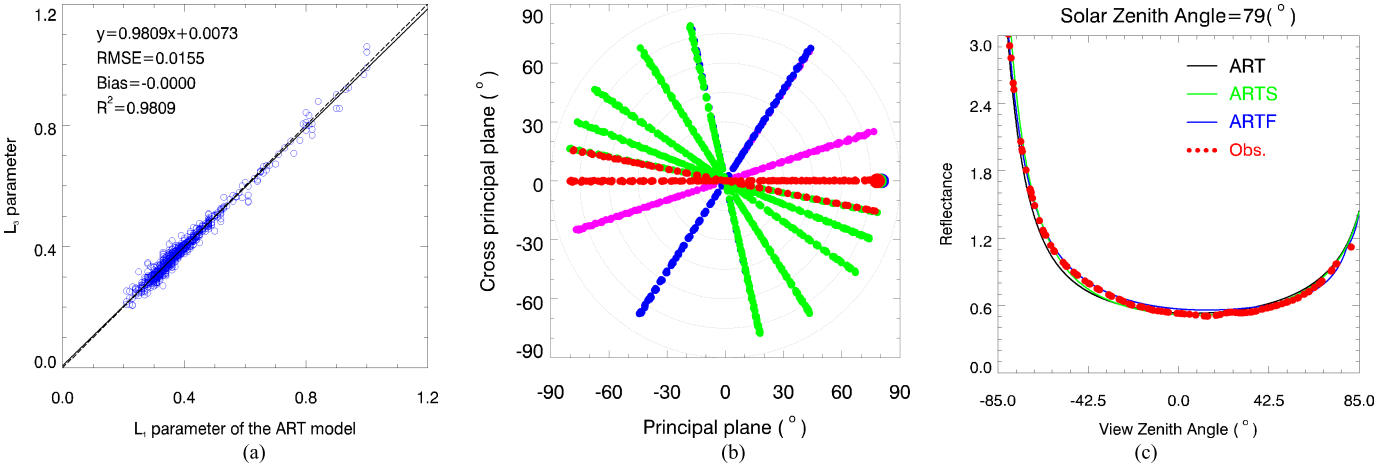


Fig. 9. (a) Scatter plot that comparing L_3 parameter for linear regression based on ARTF model parameters and L_1 parameter fitted by the ART model. (b) Angle distribution of the field-measured data. The distance between the dot and the center represents the zenith angle. The angle between the horizontal axis and the line from the center to the dot represents the azimuth angle. The larger dots represent the SZAs, and the small dots represent the VZAs. The different colors represent different SZAs and their corresponding VZAs. (c) BRDF shapes reconstructed by the ART (black), ARTS (green), and ARTF (blue) models in the PP are compared with field-measured data (red dots) in the SW-NIR (i.e., 1.02 μm). The negative axis along the PP is the forward-scattering direction, and the positive axis is the backward-scattering direction.

band of 1.02 μm since the snow parameters retrieved by the ART have a clear physical meaning. The L_2 and b parameters retrieved by the ARTF model and the L_1 parameter retrieved by the ART model have an obvious linear relationship, which can be expressed by (26). Fig. 9(a) shows the scatter plot between the L_3 and L_1 parameters, and the L_3 parameter agrees well with the L_1 parameter. The accuracy is $R^2 = 0.981$, RMSE $= 0.016$, and has a negligible bias. These findings demonstrate that the ARTF model can also be applied to retrieve snow parameters (e.g., snow grain size)

$$L_3 = 0.987L_2 - 0.224b + 0.005 \tag{26}$$

where L_2 and b are retrieved by the ARTF model.

B. Characterizing Snow BRDF Signatures at Large Angles

The measurement and modeling of snow BRDF are unavoidable problems at large angles since most of the snow is distributed in mid- to high-latitude regions. At present, the snow modeling and validation work pay more attention to small angles (i.e., SZAs $\leq 70°$ and VZAs $\leq 70°$) [11]–[13]. Therefore, we explore the performance of the ARTF model in characterizing snow BRDF signatures using field-measured data since these data can provide high-quality observations at a large angle. Fig. 9(b) shows the angle distribution of the field-measured data. The average SZA is approximately 79°, the maximum VZA can reach 82°, and the RAA can cover almost any azimuthal direction. Therefore, these typical data have sufficient observations of snow cover, which can be applied to investigate the ability of the ARTF model to characterize snow BRDF signatures at large angles. Fig. 9(c) shows the reconstructed BRDF shapes using three snow models in the PP in the SW-NIR band. It is clear that the snow reflectance has a very significant increase at large angles, especially in the forward-scattering direction, while these three models can capture this change well. These three models can perform well with respect to the characterization of snow BRDF signatures at large angles. Table VI shows that the ART and ARTS models can generally fit this typical data with R^2 values of 0.939–0.962 and RMSEs of 0.080–0.090. In comparison, the ARTF model ($R^2 = 0.985$ and RMSE $= 0.045$) agrees better than the ART and ARTS models. In addition, these three models slightly overestimate the reflectance of this typical data. In general, the RMSE values derived by these three models have increased at large angles, which still have acceptable accuracy.

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TABLE V

RETRIEVED PARAMETERS AND STATISTICAL RESULTS OF THE THREE MODELS FOR THE FIELD-MEASURED DATA IN THREE OBSERVATION DIRECTIONS' BANDS

Model	Directions	L,cm	δ	k_s	a	b	R^2	RMSE	Bias
ART	forward	0.72	--	--	--	--	0.998	0.061	-0.025
	zenith	1.00	--	--	--	--	0.999	0.016	-0.001
	backward	0.74	--	--	--	--	0.999	0.041	-0.017
ARTS	forward	1.00	0.40	0.05	--	--	0.999	0.049	-0.001
	zenith	1.00	0.30	0.10	--	--	0.998	0.021	0.004
	backward	0.99	0.91	0.44	--	--	0.999	0.033	0.000
ARTF	forward	0.96	--	--	0.30	1.10	0.999	0.017	-0.002
	zenith	0.89	--	--	0.30	0.96	0.999	0.012	-0.003
	backward	0.99	--	--	0.24	1.10	0.999	0.013	-0.002

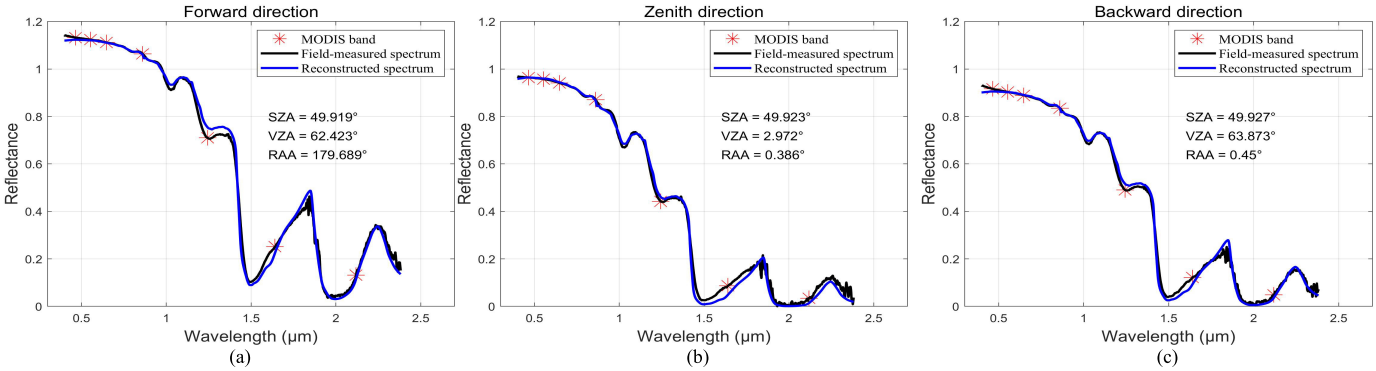


Fig. 10. Hyperspectral reflectance reconstructed (green) using the ARTF model and the hyperspectral reflectance of the field-measured data (black) for three directions [i.e., (a) forward, (b) zenith, and (c) backward directions from (Left) to (Right)]. The red asterisk represents seven bands corresponding to MODIS observations extracted from the field-measured data.

TABLE VI

RETRIEVED PARAMETERS AND STATISTICAL RESULTS OF THE THREE MODELS FOR THE FIELD-MEASURED DATA FOR THREE MODELS AT LARGE ANGLES

Model	L,cm	δ	k_s	a	b	R^2	RMSE	Bias
ART	0.47	--	--	--	--	0.939	0.090	0.008
ARTS	0.51	0.52	0.06	--	--	0.962	0.080	0.008
ARTF	0.08	--	--	0.10	0.73	0.985	0.045	0.001

C. Reconstructing Snow Hyperspectral Reflectance From Multispectral Reflectance

In this section, we investigated the ability of the ARTF model with respect to the reconstruction of snow hyperspectral from multispectral reflectance. First, we extract the seven bands corresponding to MODIS observations from the field-measured data. Then, we validate the reconstructed hyperspectral reflectance by comparison with the field-measured data.

Fig. 10 presents the hyperspectral reflectance reconstructed using the ARTF model and field-measured hyperspectral reflectance. The hyperspectral reflectance reconstructed using the ARTF model has fairly high consistency with directly fitted results (i.e., Fig. 6) by the ARTF model for three directions; the differences are negligible; and the reconstructed result of the ARTF model can generally replace the directly fitted result of the ARTF model. The hyperspectral reflectance reconstructed by the ARTF model was also shown to be quite accurate, presenting a high correlation coefficient ($R^2 = \sim 0.999$) with field-measured data (i.e., Table VII). Therefore, the ARTF model shows great potential with respect to the reconstruction of snow hyperspectral reflectance from multispectral reflectance, which is of great importance for inverting snow parameters (e.g., snow albedo and snow grain size). In addition, we pay little attention to the angle information and more attention to the spectral information in process of fitting hyperspectral data. We use the method of minimum RMSE value to determine the optimal fitting parameter values, which will lead to differences for fitting parameters in different directions. The main reason is the difference in hyperspectral reflectance in different directions. We will get the same fitting parameters when we use all the angle and spectral information. However, this is a relatively complicated process, and we do not discuss this in this article.

D. Uncertainty of Snow Data

The snow data have potential uncertainties that need to be further discussed. Global POLDER data often have a coarse spatial resolution of approximately 6×7 km [11]–[13].

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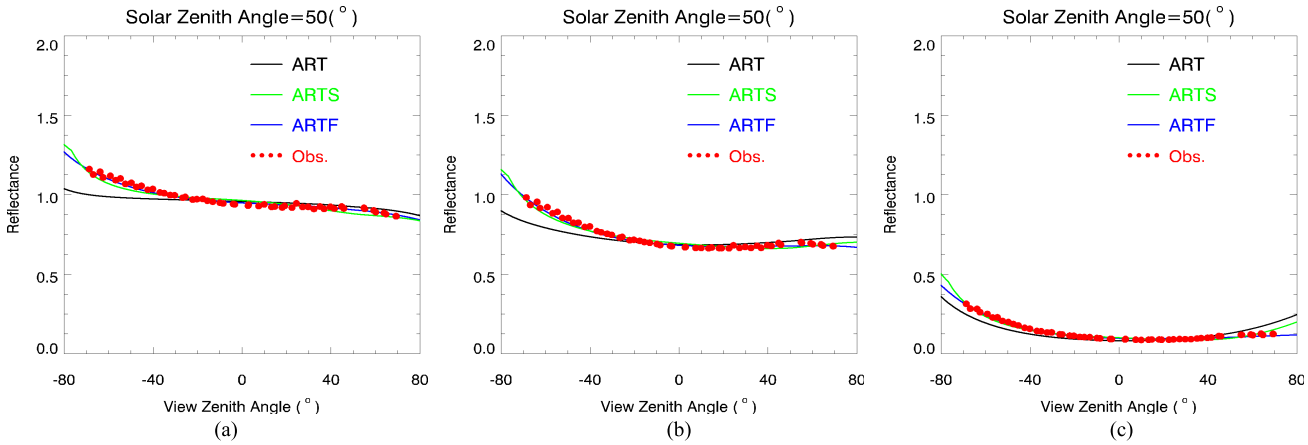


Fig. 11. BRDF shapes reconstructed by the ART (black), ARTS (green), and ARTF models (blue) in the PP compared with field-measured data (red dots). The wavelengths are (a) red, (b) SW-NIR, and (c) LW-NIR (i.e., 0.677, 1.032, and 1.649 μm, respectively). The negative axis along the PP is the forward-scattering direction, and the positive axis is the backward-scattering direction.

TABLE VII

RETRIEVED PARAMETERS AND STATISTICAL RESULTS OF THE THREE MODELS FOR THE FIELD-MEASURED DATA FOR THREE MODELS AT LARGE ANGLES

Spectrum	Directions	L,cm	a	b	R^2	RMSE	Bias
Reconstructed spectrum	forward	0.21	0.30	1.10	0.998	0.024	-0.001
	zenith	0.28	0.30	0.95	0.998	0.021	-0.009
	backward	0.37	0.15	0.96	0.998	0.018	-0.001

Therefore, we selected data with coverage of 100% to ensure homogeneity of the pixels. The shape of the CAR footprint varies with the VZA on a flat surface, and the changing footprint areas are a well-known challenge in the acquisition of multiangle measurements, which may cause uncertainties in snow BRDF data. The snow reflectance of CAR data has an obvious variation when the angle variations are small, which leads to the saturation effect for these three snow models, since CAR data have very dense BRDF sampling (see Fig. 4). This may be due to the uncertainty of the CAR data, but this has a relatively small impact on the overall fitting effect. However, this phenomenon does not exist in the LW-NIR band. The main reason is that the amount of data in the LW-NIR band (number = 203 595) is relatively small compared to the red and SW-NIR bands (number = 763 514). Therefore, the uncertainty of the CAR data does not exist in the LW-NIR band due to missing data [3]. Moreover, these three snow models can perform well in characterizing the BRDF characteristics of snow. For the field measurements, the acquired BRFs had an accuracy of 1%–5% depending on wavelength and measurement conditions, and their spectral reflectance shows a large uncertainty in the LW-NIR region [21], [27]. A local sudden break in the simulation data in the forward-scattering direction at approximately VZA $=\sim -40°$ may be caused by the interpolation procedure of the bic-PT model. However, these subtle variations in snow data should have little effect on the validation results because the snow data can generally characterize the snow hyperspectral bidirectional reflectance well. In addition, the BRDF shapes of the ARTF model have a similar result to that of the ART model over the whole hemisphere. However, the BRDF shapes of the ARTS model are significantly different from the shape of the ART model. Therefore, the ARTF model has a better correction effect on the ART model, which basically does not change the shape of the ART model. We use snow data with good BRDF sampling in this article, and the ARTS model has a significant BRDF shape distortion and causes obvious inversion errors due to a lack of observation data. The negative refraction of ice crystals that we use is derived from interpolation. Although these snow data capture uncertainties to a certain degree, such uncertainties have an equal influence on validating the ability of these three snow models. Therefore, this validation is implemented fairly and can provide convincing references for potential users.

V. CONCLUSION

In this article, we proposed a method for correcting the ART model to better model the hyperspectral bidirectional reflectance of pure snow. We comprehensively validated the ability of the corrected ART model with respect to the characterization of snow angular and spectral signatures using POLDER, CAR, field-measured, and simulating data of the bic-PT model.

Validation analyses showed that significant improvements in capturing the forward-scattering effect can be made by using the ARTF model compared with using the ART and ARTS models for satellite, aircraft, and ground data. The ARTS model can also effectively characterize snow BRDF signatures to some degree. However, the ART model tends to underestimate snow reflectance in the forward direction and overestimate snow reflectance in the backward direction. The MREs of the ARTF model are reduced by 0.9% (0.4%), 1.8% (0.5%), and 3.1% (0.1%) compared with those of the ART (ARTS) model for POLDER, CAR, and field-measured data in the red and SW-NIR bands, respectively. The ARTF model shows higher accuracy than the ART and ARTS models

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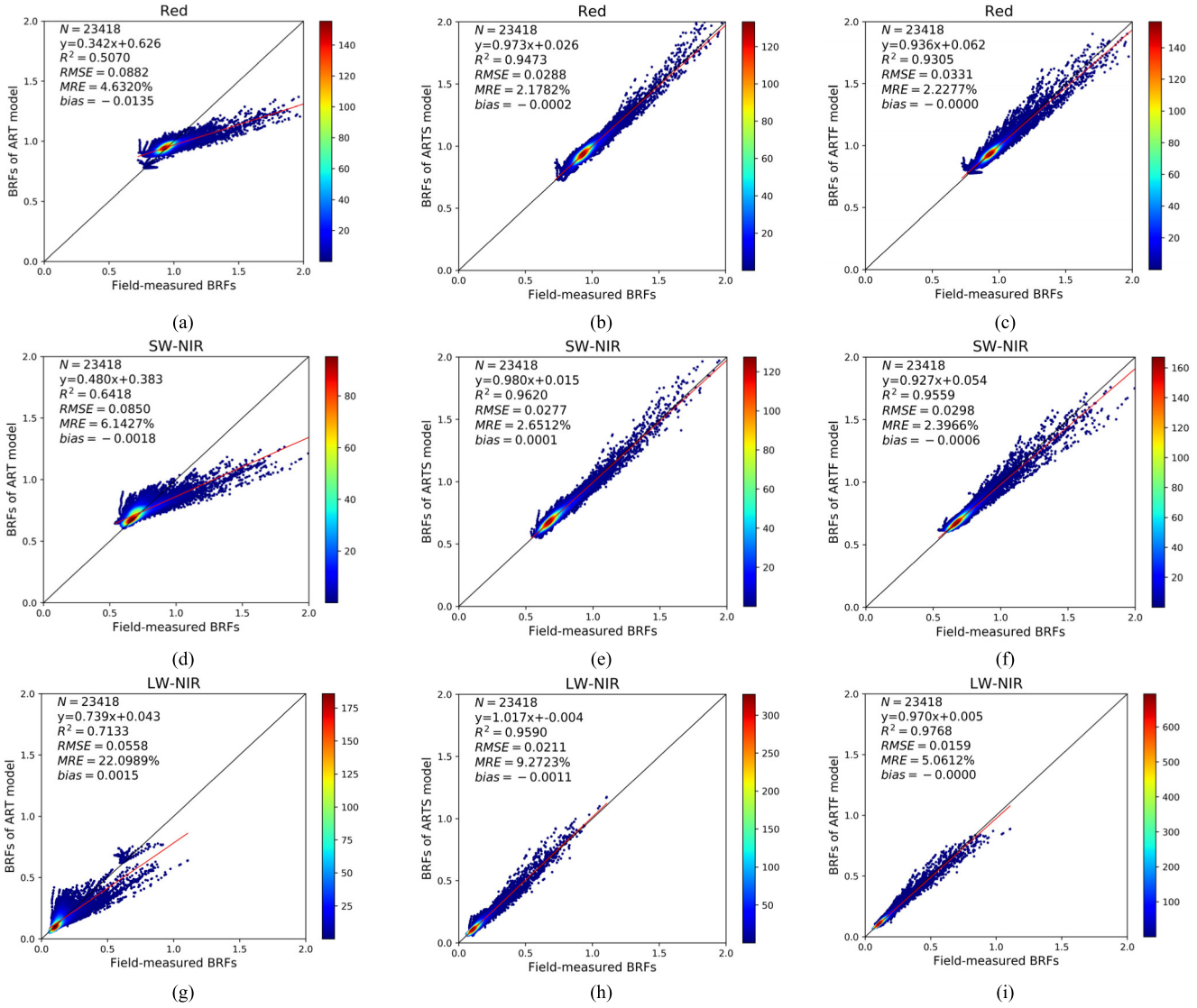


Fig. 12. Scatter plots that compare all field-measured data with the model-observation fits by the ART, ARTS, and ARTF models from (Left) to (Right) in (a)–(c) red, (d)–(f) SW-NIR, and (g)–(i) LW-NIR wavelengths (i.e., 0.677, 1.032, and 1.649 μm, respectively).

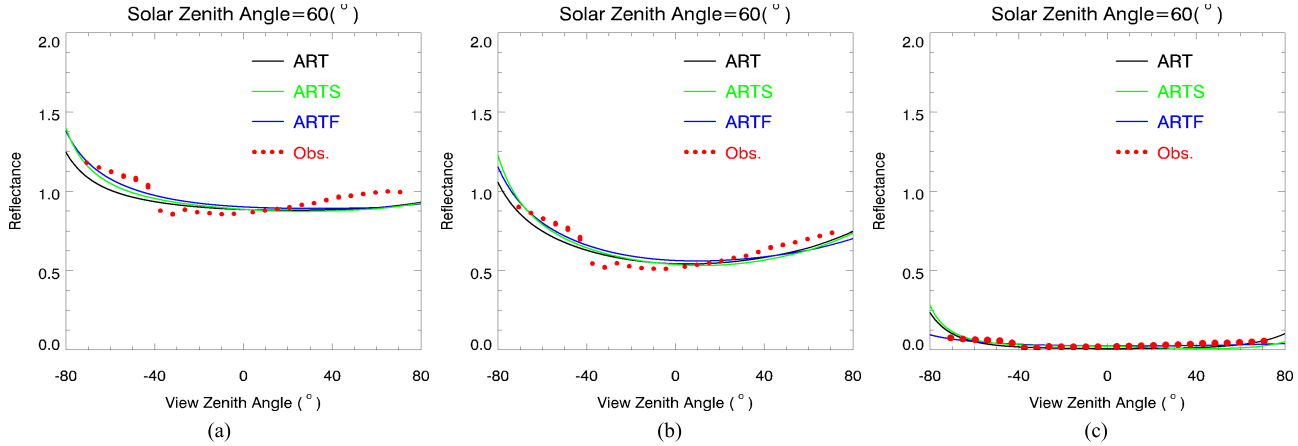


Fig. 13. BRDF shapes reconstructed by the ART (black), ARTS (green), and ARTF (blue) models in the PP compared with simulation data of the bic-PT model (red dots). The wavelengths are (a) red, (b) SW-NIR, and (c) LW-NIR (i.e., 0.67, 1.02, and 1.65 μm, respectively). The negative axis along the PP is the forward-scattering direction, and the positive axis is the backward-scattering direction.

in the LW-NIR band. The MREs of the ARTF model decreased by 21.2% (5.4%) and 17.0% (4.2%) compared to those of the ART (ARTS) model for the CAR and field-measured data, respectively. Therefore, the ARTF model has great potential to better model snow BRDF signatures, especially in the LW-NIR band. The ARTF model could capture snow hyperspectral

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TABLE VIII
BRDF PARAMETERS AND STATISTICAL RESULTS OF THE THREE MODELS FOR THE CAR DATA IN THE RED, SW-NIR, AND LW-NIR BANDS

Model	Bands	Parameters			Forward direction			Backward direction			All		
		L,cm	δ/a	k_s/b	R^2	RMSE	Bias	R^2	RMSE	Bias	R^2	RMSE	Bias
ART	red	0.24	--	--	0.745	0.42	-0.054	0.871	0.036	0.024	0.768	0.039	-0.004
	SW-NIR	0.28	--	--	0.721	0.041	-0.055	0.812	0.038	0.026	0.788	0.038	-0.001
	LW-NIR	0.19	--	--	0.687	0.034	-0.034	0.736	0.028	0.015	0.657	0.026	-0.001
ARTS	red	4.50	0.58	0.52	0.943	0.021	0.018	0.910	0.013	0.005	0.937	0.022	-0.002
	SW-NIR	0.54	0.53	0.43	0.984	0.021	0.010	0.978	0.020	0.000	0.961	0.020	-0.001
	LW-NIR	0.29	0.52	0.29	0.989	0.015	0.004	0.968	0.014	-0.001	0.962	0.014	0.001
ARTF	red	0.00	0.94	0.20	0.985	0.016	0.000	0.965	0.014	-0.003	0.970	0.016	0.003
	SW-NIR	0.02	0.77	0.26	0.991	0.010	-0.010	0.983	0.014	0.002	0.983	0.013	-0.001
	LW-NIR	0.01	0.20	0.27	0.997	0.010	-0.002	0.971	0.002	0.000	0.987	0.008	0.001

TABLE IX
BRDF PARAMETERS AND STATISTICAL RESULTS OF THE THREE MODELS FOR THE SIMULATED DATA OF THE BIC-PT MODEL IN THE RED, SW-NIR, AND LW-NIR BANDS

Model	Bands	Parameters			Forward direction			Backward direction			All		
		L,cm	δ/a	k_s/b	R^2	RMSE	Bias	R^2	RMSE	Bias	R^2	RMSE	Bias
ART	red	1.81	--	--	0.720	0.040	0.003	0.760	0.035	-0.009	0.770	0.037	-0.002
	SW-NIR	0.94	--	--	0.734	0.041	0.000	0.736	0.036	-0.011	0.725	0.038	0.005
	LW-NIR	1.06	--	--	0.645	0.025	0.018	0.687	0.023	-0.020	0.621	0.020	-0.013
ARTS	red	2.29	0.45	0.06	0.731	0.043	0.010	0.778	0.034	-0.006	0.754	0.037	0.001
	SW-NIR	1.05	0.45	0.07	0.785	0.041	0.013	0.765	0.031	-0.013	0.745	0.037	0.001
	LW-NIR	1.81	0.60	0.09	0.451	0.020	0.012	0.496	0.013	-0.004	0.412	0.012	-0.004
ARTF	red	0.45	0.95	0.06	0.761	0.037	0.006	0.787	0.032	-0.007	0.764	0.036	-0.001
	SW-NIR	0.47	0.83	0.08	0.745	0.035	0.007	0.807	0.033	-0.005	0.773	0.034	0.000
	LW-NIR	0.04	0.06	0.02	0.621	0.010	0.000	0.647	0.008	-0.002	0.608	0.007	0.000

reflectance effectively for the field-measured data. However, the ART model yielded a significant underestimation in the visible and SW-NIR regions. The ARTS model yielded a slight underestimation in the visible and SW-NIR regions and a significant overestimation in the LW-NIR region, especially at wavelengths of 1.50 and 2.00 μm. The R^2 values of these three models reached ~0.99, indicating a high correlation with field-measured data. The RMSEs of the ARTF model range from 0.018 to 0.024 and are smaller than those of the ART (RMSE = 0.025–0.059) and ARTS (RMSE = 0.025–0.031) models. The ARTF model could also effectively represent snow hyperspectral reflectance simulated using the bic-PT model with the corresponding R^2 and RMSE values reaching ~0.999 and 0.014, respectively. However, the ART and ARTS models slightly underestimated the fitting results of the hyperspectral reflectance in the visible and SW-NIR regions, and the ARTS model yielded a significant overestimation in the LW-NIR region. The R^2 and RMSE of these two models reached 0.998–0.999 and 0.021–0.061, respectively. These results demonstrate that the ARTF model can fit the simulated hyperspectral reflectance with a high degree of accuracy. In conclusion, we performed a comprehensive investigation of snow hyperspectral bidirectional reflectance using a variety of data sources, which can provide an improved understanding of the performance of these three snow models in characterizing the scattering properties of snow. We suggest further applications for the ARTF model in the near future, which can effectively capture the characteristics of snow hyperspectral bidirectional reflectance. Future work will focus on estimating snow parameters (e.g., snow albedo and snow grain size) from satellite data and further investigating the ability of the ARTF model to reconstruct snow hyperspectral reflectance from multispectral reflectance.

APPENDIX

See Figs. 11–13 and Tables VIII and IX.

ACKNOWLEDGMENT

The POLarization and Directionality of the Earth's Reflectances (POLDER)-3/Polarization and Anisotropy of

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Reflectances for Atmospheric Sciences coupled with Observations from a Lidar (PARASOL) bidirectional reflectance distribution function (BRDF) databases were developed by the Laboratoire des Sciences du Climat et de l'Environment (LSCE) and provided by the POSTEL Service Center. The authors acknowledge the use of cloud absorption radiometer (CAR) data products and imagery archived by the National Aeronautics and Space Administration (NASA)/Goddard Space Flight Center (GSFC)/Earth Science Data and Information System (ESDIS) with funding provided by NASA. They would also like to thank the three anonymous reviewers for their valuable comments and suggestions.

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